K932 — RATIFY Grace's THREE-CURVATURES distinction (extends K930 one level deeper) + refine the audit bar. There are three curvatures in the strong sector, and only ONE is the asymptotic-freedom antiscreening: (1) base $\varkappa$_Bergman = −5 (Kähler curvature of D_IV^5 → gravity a_1, F60–F66); (2) geometric CP^2 fiber = −42/25 (intrinsic curvature of the color fiber → *running corrections*, NOT the leading β_0); (3) gauge fiber F (the gluon field strength, the adjoint −2F paramagnetic, strength C_A → THE antiscreening). Aim the a_2 at the GAUGE F piece (E = −2F), not the geometric CP^2 curvature (#2) and not the base $\varkappa$ (#1). Within the Seeley-DeWitt a_4, the β-function is the gauge $tr(F^2) = tr(\Omega^2)$ piece (from E=−2F on the adjoint); the curvature pieces (base R, fiber CP^2, pure-curvature $a_4(Q^5)$=147) are SEPARATE terms (gravity/cosmological). Reproduce −11/3 = −1/3 + 4 from the GAUGE piece alone.

Keeper | 2026-07-26 Sun | Grace's catch is the correct extension of K930: base-vs-fiber wasn't fine enough — the FIBER splits into geometric-CP^2 (a correction) and gauge-F (the antiscreening). Ratified, sourced, and folded into the blind bar.

The three curvatures (ratified, Grace's sourcing)

#	curvature	value	heat-kernel role	is it the antiscreening?
1	base $\varkappa$_Bergman (Kähler, of D_IV^5)	−5	a_1 → Einstein-Hilbert (gravity, F63)	NO
2	geometric CP^2 fiber (intrinsic)	−42/25	running CORRECTIONS (curved-background)	NO (a correction, not β_0)
3	gauge fiber F (gluon field strength)	(dynamical)	a_2 gauge $tr(F^2)$, E=−2F, strength C_A	YES — the −11/3 antiscreening

- The β-function's 11/3·C_A is the gauge $tr(F^2) = tr(\Omega^2)$ piece of the a_4 Seeley-DeWitt coefficient, sourced from the adjoint endomorphism E = −2F (the gluon's color-magnetic moment). Flat-4D effect.
- The base/fiber intrinsic curvatures (#1, #2) are the pure-curvature a_4 terms (R^2, R_$\mu\nu^2$, E·R, ...) — gravity/cosmological, NOT the gauge running coefficient. Grace's $a_4(Q^5)$=147=N_c·g^2 is one of these pure-curvature terms.
- So Elie aims at E=−2F (the gauge piece). Aiming at #2 (geometric CP^2=−42/25) would give a curvature correction to the running, not β_0; aiming at #1 ($\varkappa$_Bergman) would give a gravitational term.

The deeper implication — this makes the "one operator" stretch HARDER (confirms common-cause is the bankable tier)

- Confinement (Schur/zero-Shilov, T2523) lives in the BASE (the Shilov boundary of D_IV^5 + the SU(3) center Z_3).
- Antiscreening (E=−2F, C_A) lives in the GAUGE fiber F.

- These are different curvatures / different loci (base-boundary vs gauge-fiber). So the COMMON CAUSE — the short-root structure forces non-abelian SU(3) (T666), which THEN feeds the base-confinement mechanism AND the gauge-F-antiscreening mechanism — is firmly the bankable tier.
- The ONE-OPERATOR stretch (the Schur-confinement object literally = the antiscreening-a_2 object) would have to bridge the base-Shilov and the gauge-F, which the three-curvature distinction shows are genuinely different objects. So the stretch is HARDER than it looked. Guard against anyone claiming "one operator" via a curvature conflation — it would need an explicit bridge, not a shared "3."

FF-20 quarantine — UPDATED (Grace's new trap)

The elevens (β-11 / KK dim-K=11 / Weitzenböck c_2=11 / 2n_C+1 / 137=11^2+4^2) + β_0=g=7-itself + $a_4(Q^5)$=147=N_c·g^2 (pure-curvature ≠ gauge a_2) — PLUS: - geometric CP^2 fiber curvature = −42/25 is a RUNNING CORRECTION, NOT the leading β_0. Do NOT weld it to the gauge-F antiscreening. (Grace's catch; extends the base-vs-fiber to a three-way split.)

Ratifications

- Lyra F702 correction — ACCEPTED and ratified. She took K930 in full, corrected the note in place, owns the κ_Bergman error plainly, and drew the right meta-lesson (the FF-20 knife was correct on the Jacobian [no forcing chain] and wrong on the three-hats [there IS a forcing chain] — "same discipline, opposite verdict = it's discipline, not reflex"). The corrected unification (common cause, one-operator = labeled stretch) is the honest tier.
  - HYGIENE (Lyra to fix): the note's FILENAME still says "sign_from_negative_Bergman_curvature" — the stale WRONG title contradicts the corrected content. Rename or keep a loud banner; a future reader hitting the filename could re-inherit the error. Cite the banner, not the filename.
- Grace's three-curvatures + three-hats theorem sourcing — ratified (short-root mult = n_C−2 = 3 = N_c, T666/T2511; C_A=N, center=Z_N standard → one forcing).
- Elie's adoption — ratified. Setup routes the sign through the non-abelian self-coupling, now attributed to the gauge F. Next crank = the a_2 with E=−2F on the short-root fiber, gated on Grace's book-sourced standard a_2, judged against K929/K930.

The refined blind bar (K929 + K930 + K932)

- Tier-1 WIN: the induced a_2's gauge tr(F^2) piece, from E = −2F on the geometry-forced adjoint (short roots → SU(3), C_A=3), gives β_0>0 (the +4 paramagnetic beats the −1/3 diamagnetic) — from the gauge F, target-innocently (adjoint forced by the short roots; −2F standard once adjoint). NOT from κ_Bergman (#1), NOT from geometric-CP^2 (#2).
- Reproduce −11/3 = −1/3 + 4 from the gauge piece = a consistency check the operator is aimed right, NOT a derivation of "11."
- Tier-1★ stretch: bridge base-Shilov-confinement and gauge-F-antiscreening as ONE — now shown HARDER (different curvatures); earn it explicitly.
- Tier-3: the geometric-CP^2(−42/25) / N_max / Bergman corrections → a computed MODIFICATION of the running = falsifiable (this is where #2 legitimately enters — as a correction, not β_0).

— Keeper K932, 2026-07-26. Ratify Grace's THREE curvatures: base κ_Bergman=−5 (gravity a_1) / geometric CP^2 fiber=−42/25 (running CORRECTION, not β_0) / gauge fiber F (E=−2F, C_A, THE antiscreening). Aim the a_2 at the GAUGE tr(F^2)/E=−2F piece; the curvature pieces are separate a_4 terms. Extends K930 base-vs-fiber to a 3-way split; new FF-20 quarantine: geometric-CP^2=−42/25 ≠ gauge-F. DEEPER: confinement=base-Shilov, antiscreening=gauge-F — different curvatures → the ONE-OPERATOR stretch is HARDER (needs an explicit base↔fiber bridge, not a shared 3); COMMON CAUSE (short-roots→SU(3)→both) firmly the bankable tier. Ratified: Lyra F702 correction (full accept, banner; stale filename to fix) + meta-lesson (knife right on Jacobian, wrong on three-hats); Grace three-hats theorem sourcing; Elie adoption (a_2 E=−2F, gated on Grace's standard a_2). Refined bar: sign from gauge F target-innocently = Tier-1 win; reproduce −11/3=−1/3+4 = consistency; #2 CP^2 enters only as Tier-3 correction. See [[Keeper_K930_audit_F702_confinement_AF_unification]], [[Keeper_K929_PRE_REGISTERED_a2_beta_function_audit_criteria]], [[Lyra_F702]].